WORKSHEET · SCRATCH (GR 3–5) ADVANCED: ARCADE ENGINEERS

Lesson 1: Clones: One Sprite, Many Copies

Module 1 | 45 minutes | Clones | Grades 3–5 · ages 9–11

Name: _____ Class: _____ Date: _____

Practise what you learned today. Read each task carefully.

1. Key words

Match each word to what it means. Write the letter.

a. clone ___ A. Removes a clone when it has finished.
b. original ___ B. A script that keeps making clones.
c. factory ___ C. The sprite that makes the clones.
d. delete this clone ___ D. A copy of a sprite made while the program runs.

2. Plan your code

Before you build it: Star shower: the Star hides and makes 10 clones (one every 0.3 s). Each clone starts at a random x at the top, falls to the bottom, then deletes itself. Write or draw the blocks you will use.

3. Count the copies

You click the green flag. The Star sprite runs “repeat (5) create clone of myself”. How many stars are there in total (original + clones)? Circle one.

a. A. 6
b. B. 5
c. C. 1

4. Life of a clone

Put the life of a clone in order Number them 1–5.

a. ___ The clone falls until it reaches the bottom
b. ___ The clone goes to a random place and shows itself
c. ___ The clone starts its “when I start as a clone” script
d. ___ The original runs “create clone of myself”
e. ___ The clone deletes itself

5. Plan your code

Before you build it: Snowfall forever: the Star (dressed as snow) makes a clone every 0.2 seconds, forever. Each clone falls and deletes itself at the bottom, so there are never too many. Write or draw the blocks you will use.

How did it go? Colour a face: 😀 I can do it · 🙂 Nearly there · 🤔 I need help

☐ I can make clones from a hidden original.
☐ I can give each clone its own behaviour.
☐ I can delete clones when they are done.